

99_playground

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[1]: import numpy as np
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1 University Physics with Modern Physics

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1.1 Problem 1.1 - Volume of a Cylinder

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1.1.1 Problem Statement

- Nuclear waste material in a physics lab is stored in a cylinder of height $4\frac{13}{16}$ feet and circumference $8\frac{3}{16}$ feet. What is the volume of the cylinder in metric units?

1.1.2 Given

- $h = 4.83125$ feet
- $c = 8.1875$ feet

1.1.3 Research

- Convert feet to meters
- Use the formula for the volume of a cylinder: $V = r^2h$
- To find the radius, use the formula for the circumference of a circle: $c = 2r$
- To convert feet to meters, use the conversion factor $1 \text{ foot} = 0.3048 \text{ meters}$
- Plug the values into the formulas to find the volume in cubic meters

1.1.4 Calculations

Radius:

$$r = \frac{C}{2\pi}$$

Convert to meters:

$$r_m = r \times 0.3048$$

$$h_m = h \times 0.3048$$

Volume:

$$V = \pi r^2 h$$

```
[2]: # Calculation
circumference = 8.1875
h = 4.83125
r = circumference / (np.pi * 2)
r_meters = r * 0.3048
h_meters = h * 0.3048
v = np.pi * r_meters**2*h_meters

print(f'The volume of the cylinder is {v:.2f} cubic meters.')
```

The volume of the cylinder is 0.73 cubic meters.

2 Problem 1.2 - Volume of a Barrel of Oil

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2.1 Problem Statement

The volume of a barrel of oil is 159 liters. We need to design a cylindrical container that will hold this volume of oil. The container needs to have a height of 1.00 meters to fit in a transport container. What is the required radius of the container? What is the circumference of the cylindrical container?

2.2 Given

$V = 159$ liters $h = 1.00$ meters

2.3 Research

circumference:

$$c = 2\pi r$$

Volume:

$$V = \pi r^2 h$$

2.4 Calculations

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[3]: # Calculations
V = 159
h = 1.00
r = np.sqrt(V/(np.pi*h))
c = 2 * np.pi * r
```

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print(f'The radius of the container is {r:.2f} meters.')  
print(f'The circumference of the container is {c:.2f} meters.')
```

The radius of the container is 7.11 meters.

The circumference of the container is 44.70 meters.

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